Lecture Notes: Concrete Cohomology on S^2 and T^2 from Winding Forms to Gauss–Bonnet

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1 The main question

These notes are about one question:

$$\boxed{\text{When is a closed form not exact?}}$$

That is the concrete meaning of cohomology.
We will study this through three examples:

- the winding form on $\mathbb{R}^2 \setminus \{0\}$,
- the area form on S^2,
- the forms $dx, dy, dx \wedge dy$ on T^2.

Then we will connect this to curvature and Gauss–Bonnet.

Takeaway. Cohomology measures the failure of local primitives to glue into global primitives.

2 de Rham cohomology

For a smooth manifold M,

$$H^k_{\mathrm{dR}}(M) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}.$$

So:

- ω is *closed* if $d\omega = 0$,
- ω is *exact* if $\omega = d\eta$.

Every exact form is closed:

$$\omega = d\eta \implies d\omega = d^2\eta = 0.$$

But not every closed form is exact.
That is exactly where cohomology lives.

Takeaway. A nonzero cohomology class is represented by a form that is closed but not exact.

What should I remember? To understand cohomology concretely, do not start with abstraction. Start with:

$$\text{compute } d\omega, \quad \text{compute periods/integrals}, \quad \text{test exactness}.$$

3 First example: the winding form on $\mathbb{R}^2 \setminus \{0\}$

Consider

$$\eta = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

This is the standard winding form.

3.1 Step 1: rewrite in polar coordinates

Write

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Then

$$dx = \cos\theta\, dr - r\sin\theta\, d\theta, \qquad dy = \sin\theta\, dr + r\cos\theta\, d\theta.$$

Compute:

$$\begin{aligned} -y\,dx + x\,dy &= -r\sin\theta(\cos\theta\, dr - r\sin\theta\, d\theta) + r\cos\theta(\sin\theta\, dr + r\cos\theta\, d\theta) \\ &= -r\sin\theta\cos\theta\, dr + r^2\sin^2\theta\, d\theta + r\sin\theta\cos\theta\, dr + r^2\cos^2\theta\, d\theta \\ &= r^2(\sin^2\theta + \cos^2\theta)\, d\theta \\ &= r^2\, d\theta. \end{aligned}$$

Therefore

$$\eta = d\theta.$$

So locally the winding form is just the differential of the angle.

3.2 Step 2: why it is closed

Since locally $\eta = d\theta$,

$$d\eta = d(d\theta) = 0.$$

So η is closed.

3.3 Step 3: why it is not exact

Integrate around the unit circle:

$$\gamma(t) = (\cos t, \sin t), \qquad 0 \le t \le 2\pi.$$

Then

$$\int_\gamma \eta = \int_0^{2\pi} d\theta = 2\pi.$$

But if $\eta = df$ globally, then for every closed curve,

$$\int_\gamma \eta = \int_\gamma df = 0.$$

Contradiction.
So η is not exact.
Thus

$$[\eta] \neq 0 \in H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}).$$

Takeaway. The winding form is closed but not exact because there is no globally single-valued angle function on $\mathbb{R}^2 \setminus \{0\}$.

What should I remember? The first concrete test for non-exactness is:

find a closed curve with nonzero integral.

4 Second example: the sphere S^2

Now move from a 1-form on a punctured plane to a 2-form on a sphere.

The key object is the area form.

4.1 Step 1: spherical coordinates

Write

$$X(\theta, \phi) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi), \qquad 0 < \theta < 2\pi, \quad 0 < \phi < \pi.$$

The induced metric is

$$ds^2 = d\phi^2 + \sin^2\phi\, d\theta^2.$$

So the area form is

$$dA = \sin\phi\, d\theta \wedge d\phi.$$

4.2 Step 2: why dA is closed

Since S^2 is 2-dimensional, any 3-form on S^2 is automatically zero.

But $d(dA)$ is a 3-form.

So

$$d(dA) = 0.$$

Therefore dA is closed.

4.3 Step 3: compute the total integral

Now compute:

$$\begin{aligned}
\int_{S^2} dA &= \int_0^{2\pi}\int_0^{\pi} \sin\phi\, d\phi\, d\theta \\
&= \int_0^{2\pi} [-\cos\phi]_{\phi=0}^{\phi=\pi}\, d\theta \\
&= \int_0^{2\pi} 2\, d\theta \\
&= 4\pi.
\end{aligned}$$

4.4 Step 4: why dA is not exact

If $dA = d\alpha$ globally, then by Stokes,

$$\int_{S^2} dA = \int_{S^2} d\alpha = \int_{\partial S^2} \alpha = 0.$$

But we just found

$$\int_{S^2} dA = 4\pi.$$

So dA is not exact.

Thus

$$[dA] \neq 0 \in H^2_{\mathrm{dR}}(S^2).$$

Takeaway. On the sphere, the interesting nontrivial cohomology class is in degree 2, represented by the area form.

What should I remember? For a compact oriented surface,

if a 2-form has nonzero total integral, it cannot be exact.

5 Why $H^1(S^2) = 0$

This is worth pausing over.

On $\mathbb{R}^2 \setminus \{0\}$, the hole gave a nontrivial H^1-class.

On S^2, there is no such 1-dimensional hole.

Concretely, every loop on S^2 bounds a region. So if ω is a closed 1-form and $\gamma = \partial D$, then

$$\int_\gamma \omega = \int_D d\omega = 0.$$

So all periods vanish.

Hence every closed 1-form is exact, and

$$H^1_{\mathrm{dR}}(S^2) = 0.$$

Therefore

$$H^0_{\mathrm{dR}}(S^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(S^2) = 0, \qquad H^2_{\mathrm{dR}}(S^2) \cong \mathbb{R}.$$

Takeaway. The sphere has no winding 1-classes, but it does have a nontrivial top-degree area class.

6 Local primitives on S^2

This is the sphere-version of the winding story.

Cover S^2 by

$$U_N = S^2 \setminus \{N\}, \qquad U_S = S^2 \setminus \{S\}.$$

Define

$$\alpha_N = (1 - \cos\phi)\, d\theta \qquad \text{on } U_N,$$

and

$$\alpha_S = -(1 + \cos\phi)\, d\theta \qquad \text{on } U_S.$$

Then

$$d\alpha_N = \sin\phi\, d\phi \wedge d\theta, \qquad d\alpha_S = \sin\phi\, d\phi \wedge d\theta.$$

So on both charts, the area form is locally a derivative.

Now subtract on the overlap:

$$\alpha_N - \alpha_S = (1 - \cos\phi)d\theta + (1 + \cos\phi)d\theta = 2\, d\theta.$$

So the mismatch between the two local primitives is twice the winding form.

Takeaway. The area form on S^2 is locally exact, but the local primitives differ by angular winding on the overlap.

What should I remember? For S^2:

$$\text{local exactness} + \text{overlap winding} = \text{global non-exactness}.$$

7 Third example: the torus T^2

Now we study

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2.$$

Unlike the sphere, the torus really does have two independent winding directions.

7.1 Step 1: the forms dx and dy

The coordinate forms dx and dy on $\mathbb{R}^2$ descend to global 1-forms on T^2.

They are closed:

$$d(dx) = 0, \qquad d(dy) = 0.$$

7.2 Step 2: test exactness by periods

Consider the basic loops

$$\gamma_x(t) = (t, 0) \mod \mathbb{Z}^2, \qquad \gamma_y(t) = (0, t) \mod \mathbb{Z}^2, \qquad 0 \le t \le 1.$$

Then

$$\int_{\gamma_x} dx = 1, \qquad \int_{\gamma_x} dy = 0,$$

$$\int_{\gamma_y} dx = 0, \qquad \int_{\gamma_y} dy = 1.$$

Therefore neither dx nor dy is exact.

So

$$[dx], [dy] \neq 0 \in H^1_{\mathrm{dR}}(T^2).$$

And since the two loops separate them, they are linearly independent.

Hence

$$H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2,$$

with basis $[dx], [dy]$.

Takeaway. The torus has two independent winding 1-classes: $[dx]$ and $[dy]$.

7.3 Step 3: the top class $dx \wedge dy$

Now consider

$$dx \wedge dy.$$

It is closed:

$$d(dx \wedge dy) = 0.$$

And

$$\int_{T^2} dx \wedge dy = \int_0^1 \int_0^1 dx\, dy = 1.$$

So it is not exact.

Thus

$$[dx \wedge dy] \neq 0 \in H^2_{\mathrm{dR}}(T^2).$$

So the torus has

$$H^0_{\mathrm{dR}}(T^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2, \qquad H^2_{\mathrm{dR}}(T^2) \cong \mathbb{R}.$$

Takeaway. On T^2, the top-degree class comes from wedging the two basic winding classes:

$$[dx] \wedge [dy] = [dx \wedge dy].$$

What should I remember? The torus is different from the sphere because it has genuine 1-dimensional winding directions.

8 The local-to-global point of view

Suppose ω is a closed 2-form on a manifold.

Then on each small open set U_i, the Poincaré lemma gives

$$\omega|_{U_i} = d\alpha_i.$$

So closed forms always have local primitives.

On overlaps,

$$d(\alpha_i - \alpha_j) = 0.$$

So the differences are closed 1-forms, and often locally exact:

$$\alpha_i - \alpha_j = df_{ij}.$$

These overlap functions f_{ij} measure the failure of the local primitives to glue.

Exponentiating them gives transition functions

$$g_{ij} = e^{if_{ij}},$$

which define line bundles.

Takeaway. Cohomology, overlap data, and line bundles are different languages for the same obstruction.

9 Curvature and Gauss–Bonnet

Now bring geometry into the picture.

Let M be an oriented Riemannian surface, with Gaussian curvature K and area form dA.

Then the curvature 2-form is

$$K\,dA.$$

9.1 Why this is closed

Since M is 2-dimensional,

$$d(K\,dA) = 0.$$

So $K\,dA$ defines a cohomology class in $H^2_{\mathrm{dR}}(M)$.

9.2 Why this is locally exact

Locally, if we choose a positively oriented orthonormal frame, the Levi-Civita connection is given by a local 1-form ω_{12}, and

$$d\omega_{12} = K\,dA.$$

So curvature is locally the derivative of a connection form.
Again:

local primitive exists, global primitive may fail.

That failure is the Euler class.

9.3 Gauss–Bonnet

The Gauss–Bonnet theorem says

$$\int_M K\,dA = 2\pi\chi(M).$$

So total curvature is topological.

Takeaway. The cohomology class of the curvature form remembers the Euler characteristic.

What should I remember? Gauss–Bonnet is not about the area form dA by itself. It is about the curvature form

$$K\,dA.$$

10 Check Gauss–Bonnet on S^2

For the unit sphere,

$$K \equiv 1.$$

So

$$K\,dA = dA.$$

Hence

$$\int_{S^2} K\,dA = \int_{S^2} dA = 4\pi.$$

Since

$$\chi(S^2) = 2,$$

we get

$$2\pi\chi(S^2) = 4\pi.$$

Therefore

$$\int_{S^2} K\,dA = 2\pi\chi(S^2).$$

Takeaway. On S^2, the curvature class is nontrivial:

$$[K\,dA] = [dA] \neq 0.$$

11 Check Gauss–Bonnet on T^2

Take the flat torus with metric

$$ds^2 = dx^2 + dy^2.$$

Then

$$K \equiv 0.$$

So

$$K\,dA = 0.$$

Hence

$$\int_{T^2} K\,dA = 0.$$

Since

$$\chi(T^2) = 0,$$

we get

$$\int_{T^2} K\,dA = 2\pi\chi(T^2) = 0.$$

Takeaway. On the flat torus, the curvature class vanishes even though the area class does not.

What should I remember? Do not confuse:

$$[dA] \neq 0 \qquad \text{with} \qquad [K\,dA].$$

For a flat torus,

$$[dA] \neq 0, \qquad [K\,dA] = 0.$$

12 Euler class on S^2 from stereographic coordinates

We now make the sphere story even more concrete.

Let

$$N = (0, 0, 1), \qquad S = (0, 0, -1).$$

Define stereographic coordinates:

$$z = \frac{X + iY}{1 - Z} \qquad \text{from the north pole,}$$

$$w = \frac{X - iY}{1 + Z} \qquad \text{from the south pole.}$$

On the overlap,

$$w = \frac{1}{z}.$$

So the tangent-bundle transition function is the derivative:

$$\frac{dw}{dz} = -\frac{1}{z^2}.$$

Restrict to the equator, where $z = e^{i\theta}$. Then

$$\frac{dw}{dz} = -e^{-2i\theta}.$$

This winds twice around the unit circle as θ runs from 0 to 2π.
So the tangent bundle of S^2 has clutching degree 2, hence Euler number

$$e(TS^2)[S^2] = 2.$$

This matches

$$\chi(S^2) = 2$$

and also matches

$$\frac{1}{2\pi}\int_{S^2} K\, dA = \frac{1}{2\pi} \cdot 4\pi = 2.$$

Takeaway. For the sphere, overlap winding, Euler class, and total curvature all encode the same number:

2.

13 Big picture

Let us compare the three spaces.

$\mathbb{R}^2 \setminus \{0\}$

The winding form is closed but not exact:

$$\eta = \frac{-y\, dx + x\, dy}{x^2 + y^2}, \qquad [\eta] \neq 0 \in H^1.$$

S^2

The area form is closed but not exact:

$$dA = \sin\phi\, d\theta \wedge d\phi, \qquad [dA] \neq 0 \in H^2.$$

Also,

$$\int_{S^2} K\, dA = 4\pi = 2\pi\chi(S^2).$$

T^2

The classes $[dx]$ and $[dy]$ generate H^1, and $[dx \wedge dy]$ generates H^2. For the flat torus,

$$[K\, dA] = 0.$$

Takeaway. The same idea appears everywhere:

local primitives exist, but global primitives may fail to exist.

14 Exercises

Exercise 1. *Show directly that*

$$d\left(\frac{-y\,dx + x\,dy}{x^2+y^2}\right) = 0$$

on $\mathbb{R}^2 \setminus \{0\}$.

Exercise 2. *Compute*

$$\int_{|z|=1} \frac{-y\,dx + x\,dy}{x^2+y^2}.$$

Why does this prove the winding form is not exact?

Exercise 3. *Verify that on the unit sphere,*

$$dA = \sin\phi\, d\theta \wedge d\phi.$$

Exercise 4. *Compute*

$$\int_{S^2} dA = 4\pi.$$

Why does this prove dA *is not exact?*

Exercise 5. *Compute*

$$d((1-\cos\phi)d\theta), \qquad d(-(1+\cos\phi)d\theta),$$

and compare the two local primitives on the overlap.

Exercise 6. *On* $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, *compute the periods of* dx *and* dy *on the two basic loops and prove that* $[dx]$ *and* $[dy]$ *are linearly independent.*

Exercise 7. *Compute*

$$\int_{T^2} dx \wedge dy$$

and explain why $dx \wedge dy$ *is not exact.*

Exercise 8. *Check Gauss–Bonnet directly for the unit sphere and the flat torus.*

Exercise 9. *Explain why the stereographic transition function of* TS^2 *has winding number* 2 *along the equator.*